

1.4 - Distributionally Robust Stability and the $O(1/\sqrt{n})$ Ambiguity Radius

REFLEX project - math-theory

STATUS: DONE (Priority 4, derivation/theorem). This document discharges the *mathematical* content of methodology target **1.4**. The sensitivity ε is *estimated* from a finite simulation, so the crossing $\varepsilon = \varepsilon^*$ is a random event and a bare point estimate of ε^* is not a claim. We (i) show the estimator concentrates at the parametric rate $|\hat{\varepsilon}_n - \varepsilon| = O_p(1/\sqrt{n})$ – and that this hinges on the *common-random-numbers* (CRN) construction already in `response_modulus.py`, a naive difference giving only $O(n^{-1/3})$; (ii) build an ambiguity set of radius $\delta_n = O(1/\sqrt{n})$; (iii) derive the distributionally robust stability certificate (Cao & Shi 2022) with robust boundary $\varepsilon_{\text{rob}}^* = \varepsilon^* - \delta_n$; (iv) derive the sample-complexity $n_{\text{req}} = O(\Delta^{-2})$; and (v) separate statistical from structural uncertainty.

One-line thesis. 1.1 says “stable iff $\varepsilon < \gamma/\beta$.” But you never know ε ; you know $\hat{\varepsilon}_n \pm \delta_n$ from n episodes. The honest object is the *certificate*: declare stable only when the whole ambiguity ball is on the stable side, $\hat{\varepsilon}_n + \delta_n < \gamma/\beta$. The robust boundary sits a margin $\delta_n = O(1/\sqrt{n})$ inside the nominal one, and the sample size to resolve the crossing blows up as (distance to boundary) $^{-2}$. The CRN trick in the existing probe is exactly what makes that margin shrink at $1/\sqrt{n}$ rather than $n^{-1/3}$.

1 Carry-over and new notation

From 01-analytic-stability-boundary and 03-multi-dealer-systemic-risk:

$$m = \varepsilon\beta/\gamma, \quad \varepsilon^* = \gamma/\beta \text{ (multi-dealer: } \gamma/(N_{\text{eff}}\beta)), \quad \hat{m}_n = \frac{|\text{BR}(h + \delta) - \text{BR}(h - \delta)|}{2\delta} \text{ (CRN probe).}$$

Symbol	Reading	Definition
n	episodes per estimate	<code>rrm.eval.episodes</code> / rollouts
S	seeds for cross-seed variance	§3, §8
$\hat{\varepsilon}_n$	estimator of ε from n episodes	§1 ($= \hat{m}_n \gamma/\beta$, or Sinkhorn)
σ_ε	per-episode estimator std	§2 (delta method / IPA)
δ_n	ambiguity radius	$= z \sigma_\varepsilon / \sqrt{n} = O(1/\sqrt{n})$
a, z	miscoverage; $z = z_{1-a}$	§2
U_n	ambiguity set	ε -ball or Wasserstein ball, §3
$\varepsilon_{\text{rob}}^*$	robust boundary	$= \varepsilon^* - \delta_n$, §4
Δ	distance of truth to boundary	$= \varepsilon^* - \varepsilon $, §5
n_{req}	episodes to certify at Δ	$= (z\sigma_\varepsilon/\Delta)^2$, §5
λ	Sinkhorn regularisation	§3
η_{mod}	structural relative ambiguity	§6

2 The estimation problem

The nominal theory is deterministic, $\varepsilon < \gamma/\beta$, but ε is never observed – it is estimated by the CRN BR-slope probe (or a Sinkhorn/CKS estimator) from n episodes. The probe returns the modulus $\hat{m}_n$; since γ, β are closed forms, $\hat{\varepsilon}_n = \hat{m}_n \gamma/\beta$ and $\text{Var}(\hat{\varepsilon}_n) = (\gamma/\beta)^2 \text{Var}(\hat{m}_n)$. Deciding “stable vs. unstable” is a test of $H_0 : \varepsilon \geq \varepsilon^*$ vs. $H_1 : \varepsilon < \varepsilon^*$; a point estimate without its sampling error is the overclaim this priority prevents.

3 Concentration – and why CRN is load-bearing

Smooth functional of sample averages. BR solves the FOC $F(h; \theta) = 0$ with θ the population flow moments; empirically $\hat{\text{BR}}_n = \text{BR}(\hat{\theta}_n)$, $\hat{\theta}_n = \frac{1}{n} \sum_i X_i$. As $F_h = -\gamma \neq 0$, the implicit-function theorem + CLT + delta method give $\sqrt{n}(\hat{\text{BR}}_n - \text{BR}) \rightarrow N(0, \nabla \text{BR}^\top \Sigma_X \nabla \text{BR})$.

Naive finite difference: $O(n^{-1/3})$. With *independent* noise in the two deployments,

$$\text{bias} = O(\delta^2), \quad \text{Var}(\hat{m}_n) = \frac{\sigma_{\text{BR}}^2}{2\delta^2 n} = O\left(\frac{1}{\delta^2 n}\right),$$

so $\text{MSE} = O(\delta^4) + O(1/(\delta^2 n))$ is minimised at $\delta \sim n^{-1/6}$, giving $\text{MSE} = O(n^{-2/3})$, i.e. $|\hat{m}_n - m| = O_p(n^{-1/3})$.

CRN restores $O(1/\sqrt{n})$. `response_modulus.py` drives both probes with the *same* generators, so the difference is a pathwise (IPA) derivative:

$$D(\delta, \omega) = \hat{\text{BR}}_n(h + \delta; \omega) - \hat{\text{BR}}_n(h - \delta; \omega) = 2\delta \partial_h \text{BR}(h; \omega) + O(\delta^3),$$

whence the $O(1)$ fluctuations cancel and $\text{Var}(\hat{m}_n) = \text{Var}(D)/(4\delta^2) = \sigma_{\text{IPA}}^2/n + O(\delta^2)$, which is $O(1/n)$ *uniformly in δ* . Taking δ small kills the bias for free:

$$|\hat{m}_n - m| = O_p(1/\sqrt{n}), \quad |\hat{\varepsilon}_n - \varepsilon| = O_p(1/\sqrt{n}). \quad (*)$$

The $O(1/\sqrt{n})$ radius is bought by the variance-reduction design already in the code, not assumed.

Finite-sample certificate. If the per-episode IPA derivative is sub-Gaussian (σ_ε , cap slack A4), Hoeffding gives $P(|\hat{\varepsilon}_n - \varepsilon| > t) \leq 2e^{-nt^2/(2\sigma_\varepsilon^2)}$, so w.p. $\geq 1 - a$,

$$|\hat{\varepsilon}_n - \varepsilon| \leq \delta_n := \sigma_\varepsilon \sqrt{2 \log(2/a)/n} = O(\sqrt{\log(1/a)/n}). \quad (**)$$

4 The ambiguity set

ε -ball from cross-seed variance. Run S seeds, each on n episodes: $\bar{e} = \frac{1}{S} \sum_s \hat{\varepsilon}_n^{(s)}$, $s_\varepsilon^2 = \frac{1}{S-1} \sum_s (\hat{\varepsilon}_n^{(s)} - \bar{e})^2$ estimate ε and σ_ε^2/n . Set $U_n = [\bar{e} - \delta_n, \bar{e} + \delta_n]$ with $\delta_n = z_{1-a} s_\varepsilon$ (single n -episode estimate) or $z_{1-a} s_\varepsilon / \sqrt{S}$ (pooled Sn). The $1/\sqrt{n}$ scaling is then *tested* (§8), not assumed.

Wasserstein ball / Sinkhorn rate. ε is a W_1 -Lipschitz constant, so the principled set is $U_n^W = \{Q : W_1(Q, \hat{D}_n) \leq r_n\}$. Raw empirical W_1 is cursed, $O(n^{-1/d})$; the *entropic/Sinkhorn* estimator (fixed λ) has bias $O(\lambda)$ and $O(1/\sqrt{n})$ std (Cuturi & Peyré 2019, Ch. 7), realising the $1/\sqrt{n}$ radius. By W_1 -DRO duality (worst-case risk = nominal + $r \times$ Lipschitz), an $r_n = O(1/\sqrt{n})$ ball coincides to leading order with the ε -ball; tuning λ balances bias vs. $\sigma/\sqrt{n}$.

5 The robust stability certificate

Definition 1 (robust-stable). *A configuration is robust-stable at confidence $1-a$ if $\sup_{\varepsilon' \in U_n} m(\varepsilon') = \sup_{\varepsilon' \in U_n} \varepsilon' \beta / \gamma < 1$.*

Theorem 1 (robust boundary). *With the CRN BR-slope (or fixed- λ Sinkhorn) estimator and $\delta_n = O(1/\sqrt{n})$, a configuration is robust-stable at confidence $1-a$ iff*

$$\bar{e} + \delta_n < \gamma/\beta \iff \bar{e} < \varepsilon_{\text{rob}}^* := \gamma/\beta - \delta_n.$$

The robust boundary is strictly inside the nominal one, and $\varepsilon^ - \varepsilon_{\text{rob}}^* = \delta_n = O(1/\sqrt{n}) \rightarrow 0$. The band $[\gamma/\beta - \delta_n, \gamma/\beta + \delta_n]$ is the “undecided” region.*

Proof. $m(\varepsilon') = \varepsilon' \beta / \gamma$ is affine increasing, so the sup over $U_n = [\bar{e} - \delta_n, \bar{e} + \delta_n]$ is at the upper endpoint; requiring it < 1 gives the certificate. $\varepsilon_{\text{rob}}^* < \varepsilon^*$ since $\delta_n > 0$, and the limit follows from (*). $\square$

This is Cao & Shi’s claim – the robust stable point lies inside the nominal region and widens the margin – made quantitative: the widening is δ_n , vanishing at $1/\sqrt{n}$.

6 Sample complexity

With $\Delta := |\varepsilon^* - \varepsilon|$, deciding stability at confidence $1-a$ needs $\delta_n < \Delta$, i.e.

$$n_{\text{req}}(\Delta) = (z_{1-a} \sigma_\varepsilon / \Delta)^2 = O(\Delta^{-2}). \quad (1)$$

As $\Delta \rightarrow 0$, $n_{\text{req}} \rightarrow \infty$: no finite simulation resolves the exact crossing, so a single “ $\varepsilon^* = 0.47$ ” is undefinable – only “stable / unstable / undecided at n episodes.” The predictions $n_{\text{req}} \sim \Delta^{-2}$ and $\delta_n \sim n^{-1/2}$ (log-log slope $-1/2$) are falsifiable (§8); a slope near $-1/3$ would reveal a broken CRN coupling (§3).

7 Corollaries

Corollary 1 (robust systemic boundary). *Composing with 1.3, robust-stable for N dealers iff $\bar{e} + \delta_n < \gamma/(N_{\text{eff}}\beta)$, so $\varepsilon_{\text{rob}}^* = \gamma/(N_{\text{eff}}\beta) - \delta_n$. The same absolute δ_n eats a larger fraction of the stable region as N grows, and Δ_N shrinks with N , so $n_{\text{req}} \sim \Delta_N^{-2}$ increases: systemic markets are more fragile and harder to certify.*

Corollary 2 (statistical vs. structural). δ_n is statistical (shrinks with n). Model misspecification is a fixed relative ambiguity η_{mod} ; the total set is $[\bar{e}(1 - \eta_{\text{mod}}) - \delta_n, \bar{e}(1 + \eta_{\text{mod}}) + \delta_n]$ and robust-stable iff $\bar{e}(1 + \eta_{\text{mod}}) + \delta_n < \gamma/\beta$. As $n \rightarrow \infty$, $\delta_n \rightarrow 0$ but η_{mod} remains: more simulation cannot buy away misspecification.

Corollary 3 (one-sided coverage). *By monotonicity of m in ε , the upper-confidence-bound rule has $P(\text{declare stable} \mid \text{truly unstable}) \leq a$: it never over-claims stability beyond a – the correct asymmetry for a risk statement.*

8 Validation protocol and remaining code task

Predict. From 1.1 compute $\varepsilon, \varepsilon^*, \Delta$; predict $\delta_n, \varepsilon_{\text{rob}}^*, n_{\text{req}}$. **Measure.** For a geometric grid in n and S seeds, run the probe, form $\bar{e}, s_\varepsilon, \delta_n$. **Check:** (i) s_ε vs. n log-log slope $-1/2$; (ii) certificate flips at $\varepsilon_{\text{rob}}^*$, gap collapses as $1/\sqrt{n}$; (iii) $n_{\text{req}} \sim \Delta^{-2}$. Report the phase diagram with nominal boundary, robust boundary, and shaded undecided band.

Code (out of scope here): a thin `analysis/robust_boundary.py` wrapping the existing probe in a cross-seed loop, returning $(\bar{e}, \delta_n, \varepsilon_{\text{rob}}^*, \text{band})$ and the rate check; optionally `analysis/sinkhorn_epsilon`. No change to the market model.

9 Honest caveats

- The $1/\sqrt{n}$ rate is a property of the *estimator* (CRN / fixed- λ Sinkhorn); naive difference gives $n^{-1/3}$, raw W_1 gives $n^{-1/d}$. The log-log slope check is its falsification test.
- Sub-Gaussianity assumes the cap slack (A4); deep past the boundary use a Bernstein / median-of-means variant.
- Cross-seed s_ε conflates estimator noise with reference-state drift unless γ, β, h^* are held fixed; report δ_n per reference state.
- Structural η_{mod} does not shrink; the (c_t, I) sensitivity sweep must accompany δ_n or the robustness is illusory.
- The CLT radius $(*)$ and finite-sample $(**)$ agree in rate, differ in constant; use $(**)$ for any hard certificate.

References

- T. Cao, R. Shi. *Distributionally Robust Performative Prediction*. arXiv:2206.01844, 2022. – minimax stable point; robust point inside the nominal region, widening the margin. (Lit. #13.)
- M. Cuturi, G. Peyré. *Computational Optimal Transport*. 2019 (arXiv:1803.00567), Ch. 7. – entropic/Sinkhorn estimation and its $O(1/\sqrt{n})$ sample complexity with $O(\lambda)$ bias (§4). (Lit. #14.)
- J. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative Prediction*. ICML 2020. – ε as the W_1 -Lipschitz sensitivity.
- Supporting: sub-Gaussian (Hoeffding) tail for $(**)$; delta-method / M-estimator CLT for $(*)$; Genevay et al. and Mena & Niles-Weed for the entropic-OT parametric rate.
- Builds on 01-analytic-stability-boundary (closed forms + CRN probe) and 03-multi-dealer-systemic-r (§6.1).